

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 6 — Electrochemistry

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	6.1 Redox Processes · Oxidation Numbers · Disproportionation · Titrations · 6.2 Electrolysis & Faraday's Law
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	246 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

6.1 Redox processes

— Rules for oxidation states, balancing complex ionic half-equations, disproportionation reactions, s

6.2 Electrolysis— Electrolysis of molten salts (PbBr₂, Al₂O₃) and aqueous solutions (brine, CuSO₄, H₂SO₄), preferen

SUBTOPIC 6.1 — REDOX PROCESSES, OXIDATION NUMBERS & TITRATIONS

1 Assigning Oxidation Numbers in Sulfur Compounds — 9701/22/M/J/21/Q4(a)–(c)

EASY · Syllabus 6.1

Sulfur forms a wide variety of compounds exhibiting different oxidation numbers.

(a) State the oxidation number of sulfur in each of the following species:

[4]

- (i) H_2S
- (ii) S_8
- (iii) SO_2
- (iv) SO_4^{2-}

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(b) Determine the oxidation number of sulfur in the thiosulfate ion, $\text{S}_2\text{O}_3^{2-}$, and in the tetrathionate ion, $\text{S}_4\text{O}_6^{2-}$.

[2]

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2 Disproportionation of Chlorine in Aqueous Alkali — 9701/21/O/N/20/Q4(a)–(d)

HARD · Syllabus 6.1

Chlorine reacts with aqueous sodium hydroxide under different temperature conditions as illustrated in Fig. 6.1.

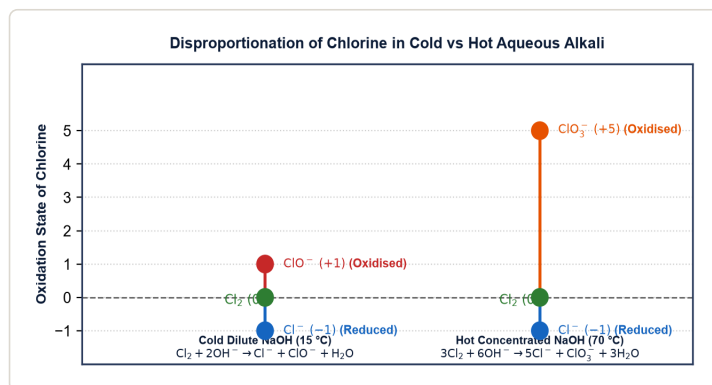


Fig. 6.1: Disproportionation oxidation states of chlorine in cold vs hot aqueous alkali.

(a) Define the term disproportionation.

[2]

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(b) Write the balanced chemical equation for the reaction of chlorine with cold, dilute aqueous sodium hydroxide (15 °C).

[2]

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(c) State the oxidation number of chlorine in each product formed in (b).

[2]

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(d) Write the balanced chemical equation for the reaction of chlorine with hot, concentrated aqueous sodium hydroxide (70 °C) and deduce the oxidation state of chlorine in the chlorate(V) product.

[2]

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3 Balancing Redox Half-Equations in Acidic Solution — 9701/23/M/J/22/Q4(a)-(b)

EASY · Syllabus 6.1

Acidified potassium dichromate(VI) is a powerful oxidising agent used in organic and analytical chemistry.

- (a) Construct the balanced ionic half-equation for the reduction of dichromate(VI) ions, $\text{Cr}_2\text{O}_7^{2-}$, to chromium(III) ions, Cr^{3+} , in acidic solution. [2]

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- (b) Construct the overall balanced ionic equation for the oxidation of iron(II) ions, Fe^{2+} , by acidified $\text{Cr}_2\text{O}_7^{2-}$. [2]

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- (c) State the colour change observed during this reaction. [1]

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4 Redox Titration: Potassium Manganate(VII) & Ethanedioate — 9701/22/F/M/21/Q5(a)-(c)

HARD · Syllabus 6.1

Standard potassium manganate(VII), KMnO_4 , is titrated against warm acidified ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$ (shown in Fig. 6.2):

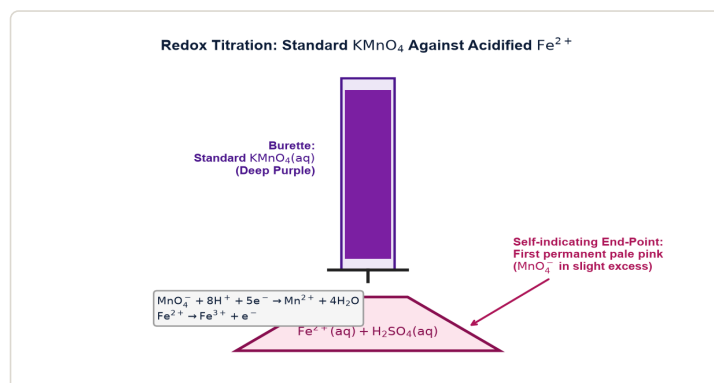
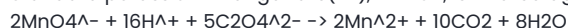


Fig. 6.2: Setup for self-indicating redox titration of acidified Fe^{2+} or $\text{C}_2\text{O}_4^{2-}$ with KMnO_4 .

- (a) State the oxidation number change of carbon in this reaction. [1]

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- (b) A 25.0 cm³ sample of 0.0500 mol dm⁻³ $\text{H}_2\text{C}_2\text{O}_4$ required 20.40 cm³ of $\text{KMnO}_4(\text{aq})$ for complete oxidation. Calculate the concentration of the KMnO_4 solution in mol dm⁻³. [3]

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- (c) Explain why no external chemical indicator is required for this titration. [1]

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5 Iodine–Thiosulfate Titration: Analysis of Bleach — 9701/21/M/J/22/Q4(a)–(c)

HARD · Syllabus 6.1

The active ingredient in commercial household bleach is the chlorate(I) ion, ClO^- .

Stage 1: $\text{ClO}^- + 2\text{I}^- + 2\text{H}^+ \rightarrow \text{I}_2 + \text{Cl}^- + \text{H}_2\text{O}$

Stage 2: $\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$

- (a) Identify the reducing agent in Stage 1 and the oxidising agent in Stage 2. [2]

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- (b) State the indicator used for the iodine–thiosulfate titration in Stage 2 and state the colour change observed at the end-point. [2]

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- (c) A 10.0 cm³ sample of bleach was diluted to 250 cm³. A 25.0 cm³ aliquot of this diluted solution was reacted with excess acidified KI and titrated with 22.80 cm³ of 0.100 mol dm⁻³ Na₂S₂O₃. Calculate the concentration of ClO^- in the original undiluted bleach in mol dm⁻³. [3]

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6 Dual Redox Nature of Hydrogen Peroxide, H₂O₂ — 9701/22/O/N/23/Q4(a)–(c)

HARD · Syllabus 6.1

In hydrogen peroxide, H₂O₂, oxygen has an unusual oxidation number of -1. H₂O₂ can act as an oxidising agent, a reducing agent, or undergo disproportionation.

- (a) Write the balanced equation for the catalytic disproportionation of H₂O₂ into water and oxygen, and show the oxidation numbers of oxygen throughout. [2]

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- (b) When H₂O₂ acts as a reducing agent in reaction with acidified KMnO₄, oxygen gas is evolved. Write the oxidation half-equation for H₂O₂. [1]

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- (c) When H₂O₂ acts as an oxidising agent in reaction with acidified Fe²⁺, write the reduction half-equation for H₂O₂. [1]

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7 Oxidation Numbers in Nitrogen Compounds — 9701/21/M/J/23/Q5(a)–(b)

EASY · Syllabus 6.1

Nitrogen exhibits every integral oxidation state from -3 to +5 across its various compounds.

- (a) Determine the oxidation number of nitrogen in each of the following: [5]

- (i) Ammonia, NH₃
- (ii) Hydrazine, N₂H₄
- (iii) Dinitrogen monoxide, N₂O
- (iv) Nitrogen dioxide, NO₂
- (v) Nitrate ion, NO₃⁻

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8 Copper Reaction with Dilute vs Concentrated Nitric Acid — 9701/22/M/J/20/Q5(a)–(c)

HARD · Syllabus 6.1

Copper metal reacts differently with concentrated and dilute nitric acid.

Reaction 1 (concentrated): $\text{Cu(s)} + 4\text{HNO}_3(\text{aq}) \rightarrow \text{Cu(NO}_3)_2(\text{aq}) + 2\text{NO}_2(\text{g}) + 2\text{H}_2\text{O(l)}$

Reaction 2 (dilute): $3\text{Cu(s)} + 8\text{HNO}_3(\text{aq}) \rightarrow 3\text{Cu(NO}_3)_2(\text{aq}) + 2\text{NO(g)} + 4\text{H}_2\text{O(l)}$

(a) Identify which species is oxidised and which species is reduced in Reaction 1. **[2]**

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(b) Construct the ionic half-equation for the reduction of nitrate ions, NO_3^- , to nitrogen monoxide, NO, in acidic solution (Reaction 2). **[2]**

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(c) State the observation that confirms nitrogen dioxide, NO_2 , is evolved in Reaction 1. **[1]**

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9 Disproportionation of Copper(I) in Aqueous Solution — 9701/21/O/N/22/Q5(a)–(b)

EASY · Syllabus 6.1

When solid copper(I) oxide, Cu_2O , is reacted with dilute sulfuric acid, a brown precipitate of copper metal and a blue solution of copper(II) sulfate are formed:

$\text{Cu}_2\text{O(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Cu(s)} + \text{CuSO}_4(\text{aq}) + \text{H}_2\text{O(l)}$

(a) Write the ionic equation for the disproportionation of aqueous copper(I) ions. **[1]**

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(b) Explain why this reaction is classified as a disproportionation by referring to changes in oxidation numbers. **[2]**

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10 Reaction of Acidified Dichromate(VI) with Sulfur Dioxide — 9701/23/O/N/21/Q4(a)–(c)

HARD · Syllabus 6.1

Sulfur dioxide gas acts as a reducing agent when bubbled through acidified potassium dichromate(VI) solution.

(a) Construct the oxidation half-equation for SO_2 in aqueous solution forming sulfate ions, SO_4^{2-} . **[2]**

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(b) Combine your half-equation in (a) with the reduction half-equation for $\text{Cr}_2\text{O}_7^{2-}$ to write the overall balanced ionic equation. **[2]**

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(c) State the visible colour change that serves as a diagnostic test for sulfur dioxide gas. **[1]**

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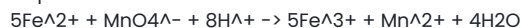
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Redox Titration: Determination of Iron in Dietary Tablets — 9701/22/F/M/23/Q4(a)–(c)

HARD · Syllabus 6.1

Iron deficiency tablets contain hydrated iron(II) sulfate, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ ($M_r = 277.9$).

Five crushed tablets with a total mass of 1.750 g were dissolved in dilute H_2SO_4 to make 250.0 cm^3 of solution. A 25.0 cm^3 sample required 18.20 cm^3 of 0.0100 mol dm^{-3} KMnO_4 for complete titration:



- (a) Calculate the amount, in moles, of Fe^{2+} ions present in the 25.0 cm^3 sample. [2]

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- (b) Calculate the total mass of Fe ($A_r = 55.8$) in the five tablets. [2]

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- (c) Calculate the percentage by mass of iron, Fe, in the tablets. [1]

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12

Oxidation States of Vanadium Oxoanions — 9701/21/M/J/21/Q6(a)–(b)

HARD · Syllabus 6.1

Vanadium forms oxoanions with characteristic colours corresponding to oxidation states +5, +4, +3, and +2:

- VO_2^+ (yellow): oxidation state +5
- VO^{2+} (blue): oxidation state +4
- V^{3+} (green): oxidation state +3
- V^{2+} (violet): oxidation state +2

- (a) Verify the oxidation state of vanadium in VO_2^+ and in VO^{2+} by calculation. [2]

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- (b) When zinc metal is added to an acidified solution of VO_2^+ , the solution gradually turns green, then blue, and finally violet. Write the ionic equation for the first reduction step from VO_2^+ to VO^{2+} . [2]

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13

Halogen Displacement Reactions & Oxidising Strength — 9701/22/M/J/22/Q5(a)–(b)

EASY · Syllabus 6.1

Halogens act as oxidising agents by accepting electrons to form halide ions:



- (a) State the trend in oxidising ability of the halogens down Group 17 from chlorine to iodine. [1]

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- (b) Write the ionic equation for the reaction that occurs when chlorine water is added to aqueous potassium bromide, and state the colour observed. [2]

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- (c) Explain the trend in oxidising ability down Group 17 in terms of atomic radius and electron shielding. [2]

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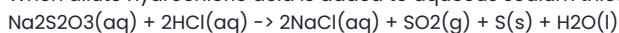
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14 Acid Disproportionation of Sodium Thiosulfate — 9701/22/O/N/21/Q6(a)–(c)

EASY · Syllabus 6.1

When dilute hydrochloric acid is added to aqueous sodium thiosulfate, the solution gradually turns cloudy yellow:



- (a) Identify the substance responsible for the yellow turbidity. [1]

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- (b) Show, by assigning oxidation numbers, that this is a disproportionation reaction of sulfur. [3]

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15 Oxidation States of Carbon in Organic Molecules — 9701/21/O/N/23/Q4(a)–(b)

HARD · Syllabus 6.1

Assigning oxidation numbers to carbon atoms reveals whether an organic transformation is an oxidation, reduction, or substitution.

- (a) Assign the average oxidation number of carbon in each of the following molecules: [5]

- (i) Methane, CH_4
 (ii) Methanol, CH_3OH
 (iii) Methanal, HCHO
 (iv) Methanoic acid, HCOOH
 (v) Carbon dioxide, CO_2

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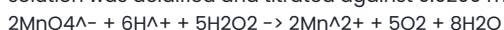
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16 Redox Titration: Purity of Hydrogen Peroxide — 9701/23/M/J/20/Q4(a)–(c)

HARD · Syllabus 6.1

A 20.0 cm³ sample of commercial hydrogen peroxide was diluted to 500 cm³ in a volumetric flask. A 25.0 cm³ portion of this diluted solution was acidified and titrated against 0.0200 mol dm⁻³ KMnO_4 :



The mean titre was 18.50 cm³.

- (a) Calculate the amount, in moles, of MnO_4^- used in the titration. [1]

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- (b) Calculate the concentration, in mol dm⁻³, of H_2O_2 in the original commercial solution. [3]

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17 Mohr's Salt Redox Titration with Dichromate(VI) — 9701/22/F/M/24/Q4(a)–(b)

HARD · Syllabus 6.1

Mohr's salt, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$ ($M_r = 392.1$), is used as a primary standard for Fe^{2+} .

- (a)** Explain why Mohr's salt is preferred over iron(II) sulfate crystals, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, as a primary standard in analytical laboratories.

[1]

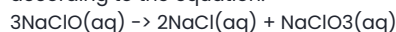
- (b)** A 3.921 g sample of Mohr's salt was dissolved and made up to 100 cm³ with dilute sulfuric acid. Calculate the volume of 0.0200 mol dm⁻³ $\text{K}_2\text{Cr}_2\text{O}_7$ required to titrate a 25.0 cm³ aliquot of this solution.

[3]

18 Thermal Disproportionation of Bleach — 9701/21/M/J/24/Q5(a)–(b)

HARD · Syllabus 6.1

When an aqueous solution of sodium chlorate(I), NaClO , is stored at elevated temperatures (above 40 °C), it slowly degrades according to the equation:



- (a)** Deduce the oxidation state of chlorine in each of the three chlorine-containing compounds.

[3]

- (b)** Explain why bleach loses its germicidal disinfectant power when heated.

[1]

19 Halide Reactions with Concentrated Sulfuric Acid — 9701/22/O/N/24/Q5(a)–(d)

HARD · Syllabus 6.1

When solid sodium halides are treated with concentrated sulfuric acid, a mixture of acid–base and redox reactions occurs depending on the reducing power of the halide.

- (a)** Explain why NaCl reacts with concentrated H_2SO_4 to produce only steamy acidic fumes of HCl with no redox reaction.

[1]

- (b)** State the observation and write the balanced redox equation for the reduction of H_2SO_4 by sodium bromide, NaBr .

[2]

- (c)** When sodium iodide, NaI , reacts with concentrated H_2SO_4 , three different reduction products of sulfur are formed: SO_2 , S , and H_2S . State the oxidation number of sulfur in each product.

[3]

20 Balancing Redox Reactions in Alkaline Solution — 9701/21/O/N/24/Q5(a)–(b)

HARD · Syllabus 6.1

In neutral or alkaline solution, manganate(VII) ions are reduced to a dark brown precipitate of manganese(IV) oxide, MnO_2 :
 $\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^-$

- (a) Construct the oxidation half-equation for the oxidation of sulfite ions, SO_3^{2-} , to sulfate ions, SO_4^{2-} , in alkaline solution. [2]

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- (b) Deduce the overall balanced ionic equation for the reaction between MnO_4^- and SO_3^{2-} in alkaline solution. [2]

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21 Disproportionation of Nitrous Acid, HNO_2 — 9701/23/M/J/23/Q4(a)–(b)

EASY · Syllabus 6.1

Nitrous acid, HNO_2 , is unstable at room temperature and disproportionates:
 $3\text{HNO}_2(\text{aq}) \rightarrow \text{HNO}_3(\text{aq}) + 2\text{NO}(\text{g}) + \text{H}_2\text{O}(\text{l})$

- (a) Determine the oxidation state of nitrogen in HNO_2 , HNO_3 , and NO . [3]

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- (b) State how this reaction demonstrates disproportionation. [1]

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22 Determination of Copper in Brass Alloy via Redox Titration — 9701/22/M/J/25/Q3(a)–(c)

HARD · Syllabus 6.1

Brass is an alloy of copper and zinc.

A 2.500 g sample of brass was dissolved in concentrated HNO_3 , neutralised, and made up to 250.0 cm^3 . Excess KI was added to a 25.0 cm^3 aliquot:



The liberated I_2 required 23.60 cm^3 of 0.100 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$ for complete titration.

- (a) State the oxidation number of copper in the precipitate CuI . [1]

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- (b) Calculate the mass of copper ($A_r = 63.5$) present in the 2.500 g brass sample. [3]

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- (c) Calculate the percentage of copper by mass in this brass alloy. [1]

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23 Unusual Oxidation States of Oxygen — 9701/21/F/M/25/Q4(a)–(b)

EASY · Syllabus 6.1

Although oxygen almost always displays an oxidation number of –2, exceptional states exist.

(a) State the oxidation number of oxygen in each of the following:

[4]

- (i) H_2O
- (ii) H_2O_2
- (iii) KO_2
- (iv) OF_2

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(b) Explain why oxygen has a positive oxidation number in oxygen difluoride, OF_2 .

[1]

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24 Comproportionation of Iodate(V) with Iodide — 9701/22/O/N/25/Q5(a)–(b)

HARD · Syllabus 6.1

When potassium iodate(V), KIO_3 , reacts with potassium iodide in the presence of dilute sulfuric acid, elemental iodine is produced:

$$\text{IO}_3^-(\text{aq}) + 5\text{I}^-(\text{aq}) + 6\text{H}^+(\text{aq}) \rightarrow 3\text{I}_2(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$$

(a) Explain why this reaction is described as a comproportionation reaction (the reverse of disproportionation).

[2]

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(b) Deduce the molar ratio of oxidising agent to reducing agent in this reaction.

[1]

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25 Sequential Reduction of Vanadate(V) by Zinc — 9701/23/O/N/24/Q4(a)–(c)

HARD · Syllabus 6.1

When granulated zinc is added to an acidified yellow solution of ammonium vanadate, NH_4VO_3 , the solution undergoes dramatic color transitions:

Yellow (+5) → Blue (+4) → Green (+3) → Violet (+2)

(a) Identify the vanadium species responsible for the blue colour and the violet colour.

[2]

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(b) Write the balanced ionic equation for the reduction of the blue species to the green species by zinc.

[2]

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26 Interference of Chloride Ions in Permanganate Titrations — 9701/21/M/J/25/Q6(a)–(b)

HARD · Syllabus 6.1

When iron(II) solutions are titrated with standard KMnO_4 , dilute sulfuric acid must be used for acidification. Dilute hydrochloric acid cannot be used.

- (a) Explain why hydrochloric acid is unsuitable for acidifying potassium manganate(VII) titrations. [2]
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- (b) Explain how the use of HCl(aq) would affect the calculated concentration of Fe^{2+} . [2]
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SUBTOPIC 6.2 — ELECTROLYSIS & QUANTITATIVE FARADAY CALCULATIONS

27 Electrolysis of Concentrated Aqueous Sodium Chloride (Brine) — 9701/22/M/J/21/Q5(a)-(d)

HARD · Syllabus 6.2

The industrial chlor-alkali process involves the electrolysis of concentrated aqueous NaCl (brine), shown in Fig. 6.3.

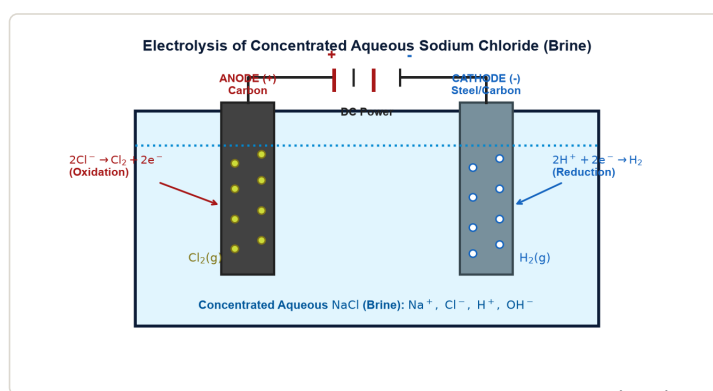


Fig. 6.3: Electrolytic cell for concentrated aqueous sodium chloride (brine).

- (a) List all four ions present in concentrated aqueous sodium chloride solution. [2]
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- (b) Write the ionic half-equation for the reaction occurring at the anode (+) and explain why chlorine is discharged rather than oxygen. [2]
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- (c) Write the ionic half-equation for the reaction occurring at the cathode (-). [1]
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- (d) Identify the alkaline chemical product remaining in the solution after prolonged electrolysis. [1]
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28 Industrial Electrolytic Refining of Copper — 9701/21/O/N/21/Q5(a)-(d)

EASY · Syllabus 6.2

Impure blister copper is purified to 99.99% purity using the electrolytic cell shown in Fig. 6.4.

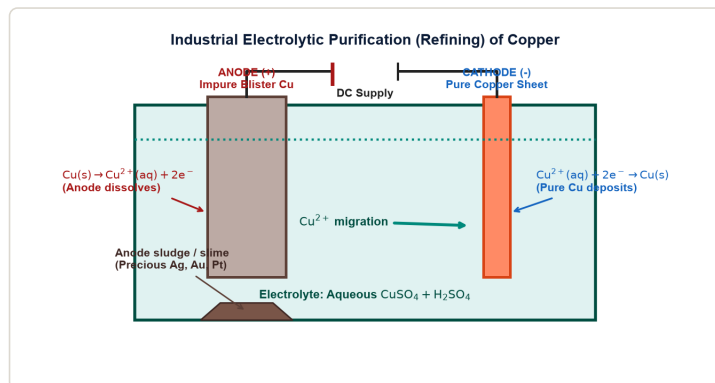


Fig. 6.4: Industrial electrolytic cell for the refining and purification of copper.

- (a) State the material used for the anode and the material used for the cathode. [2]
- (b) Write half-equations for the reactions occurring at the anode and cathode. [2]
- (c) Explain why the concentration of Cu^{2+} ions in the electrolyte remains virtually constant throughout the process. [1]
- (d) Explain why valuable precious metals like silver and gold accumulate in the anode sludge at the bottom of the cell. [2]

29 Fundamental Principles of Electrolysis — 9701/23/M/J/22/Q5(a)–(b)

EASY · Syllabus 6.2

Electrolysis is the decomposition of an electrolyte by the passage of direct electric current.

- (a) Define the term electrolyte. [2]
- (b) Distinguish between how electrical current is conducted through a metallic wire compared to through an electrolyte. [2]

30 Electrolysis of Molten Lead(II) Bromide — 9701/22/F/M/22/Q4(a)–(c)

EASY · Syllabus 6.2

When solid lead(II) bromide, PbBr_2 , is heated until molten, it conducts electricity and undergoes electrolysis using graphite electrodes.

- (a) Explain why solid PbBr_2 does not conduct electricity, but molten PbBr_2 does. [2]

- (b) Write half-equations with state symbols for the reactions at the anode and cathode. [2]
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- (c) State one observation at the anode and one observation at the cathode. [2]
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31 Electrolysis of Dilute vs Concentrated Aqueous NaCl — 9701/21/M/J/22/Q5(a)–(b)

HARD · Syllabus 6.2

The concentration of halide ions in aqueous solutions influences which anion is discharged at the inert anode.

- (a) Identify the gas evolved at the anode during the electrolysis of very dilute aqueous sodium chloride, and write the half-equation for its formation. [2]
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- (b) Explain why changing to concentrated aqueous sodium chloride changes the primary product formed at the anode. [2]
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32 Faraday's Laws & Calculation of Deposited Mass — 9701/22/O/N/22/Q4(a)–(c)

EASY · Syllabus 6.2

Faraday's constant, F , is the quantity of electric charge carried by one mole of electrons: $F = 96500 \text{ C mol}^{-1}$.

- (a) State the mathematical relationship connecting electrical charge Q , current I , and time t . [1]
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- (b) A steady electric current of 2.50 A is passed through a solution of copper(II) sulfate for 45.0 minutes. Calculate the total charge Q passed in Coulombs. [2]
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- (c) Calculate the mass of copper ($A_r = 63.5$) deposited at the cathode. [2]
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33 Determining Avogadro Constant from Electrolytic Deposition of Silver — 9701/23/O/N/22/Q4(a)–(c)

HARD · Syllabus 6.2

In an electrochemical experiment to evaluate the Avogadro constant, L , a current of 0.400 A was passed through an aqueous silver nitrate solution for 1.00 hour. The cathode gained 1.610 g of pure silver ($A_r = 107.9$). The charge on a single electron is $e = 1.602 \times 10^{-19} \text{ C}$.

- (a) Calculate the total electric charge passed during the experiment. [1]
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- (b) Calculate the experimental value for the Faraday constant, F . [2]

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- (c) Using $F = L \times e$, calculate the experimental value for the Avogadro constant, L , to 3 significant figures. [2]

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34 Electrolysis of Aqueous CuSO_4 with Inert vs Active Electrodes — 9701/22/F/M/23/Q5(a)–(b)

HARD · Syllabus 6.2

Aqueous copper(II) sulfate is electrolysed in two separate experiments:

- Cell 1: Using inert platinum electrodes
- Cell 2: Using active copper electrodes

- (a) For Cell 1 (platinum electrodes), state the product formed at the anode and write the half-equation for its formation. [2]

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- (b) State and explain the change in color of the electrolyte in Cell 1 and in Cell 2 after 30 minutes of electrolysis. [3]

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35 Electrolysis of Dilute Sulfuric Acid (Hoffman Voltameter) — 9701/21/M/J/23/Q6(a)–(c)

EASY · Syllabus 6.2

Dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, is electrolysed in a Hoffman voltameter using inert platinum electrodes.

- (a) Write ionic half-equations for the reactions occurring at the cathode and anode. [2]

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- (b) State the theoretical ratio of the volume of gas collected at the cathode to the volume of gas collected at the anode. [1]

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- (c) Explain this volume ratio in terms of the number of electrons transferred. [2]

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36 Preferential Discharge of Anions — 9701/22/M/J/23/Q4(b)

EASY · Syllabus 6.2

The ease of discharge of anions at the anode during electrolysis depends on standard electrode potentials and ionic concentrations.

- (a) Arrange the following anions in order of increasing ease of discharge at an inert anode: [2]

OH^- , I^- , Cl^- , SO_4^{2-}

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- (b) Explain why sulfate ions, SO_4^{2-} , and nitrate ions, NO_3^- , are never discharged from aqueous solutions at an inert anode. [2]
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37 Electroplating of Nickel onto a Steel Spoon — 9701/21/O/N/23/Q5(a)–(c)

EASY · Syllabus 6.2

A steel spoon is electroplated with nickel to prevent corrosion and improve appearance.

- (a) State whether the steel spoon should be connected to the positive or negative terminal of the DC power supply. [1]
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- (b) Suggest a suitable electrolyte and anode material for this electroplating process. [2]
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- (c) Write the half-equation for the nickel coating process occurring on the spoon. [1]
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38 Anodising of Aluminium — 9701/22/O/N/23/Q5(b)

HARD · Syllabus 6.2

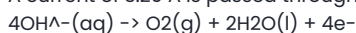
Aluminium objects are anodised to increase the thickness of the protective surface layer of aluminium oxide, Al_2O_3 .

- (a) Explain why the aluminium object is made the anode rather than the cathode in this process. [2]
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- (b) Write the overall ionic equation for the formation of the aluminium oxide layer at the anode using dilute sulfuric acid as the electrolyte. [2]
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- (c) State one property imparted to aluminium by anodising. [1]
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39 Quantitative Electrolysis: Volume of Oxygen Evolved — 9701/23/M/J/24/Q5(a)–(b)

HARD · Syllabus 6.2

A current of 3.20 A is passed through dilute sodium hydroxide for 25.0 minutes using inert electrodes.



- (a) Calculate the amount, in moles, of electrons transferred. [2]
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- (b) Calculate the volume, in cm^3 , of oxygen gas evolved at room temperature and pressure (RTP, molar volume = $24.0 \text{ dm}^3 \text{ mol}^{-1}$). [2]

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40 Extraction of Aluminium: Hall-Héroult Process — 9701/21/M/J/24/Q6(a)-(c)

HARD · Syllabus 6.2

Aluminium is extracted industrially by the electrolysis of molten alumina, Al_2O_3 , dissolved in molten cryolite, Na_3AlF_6 , at 950°C .

- (a) State two reasons why molten cryolite is added to the alumina. [2]

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- (b) Write the half-equations for the cathode and anode reactions. [2]

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- (c) Explain why the carbon anodes must be regularly replaced in this industrial process. [2]

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41 Selective Discharge of Cations in Aqueous Electrolysis — 9701/22/M/J/24/Q6(b)

EASY · Syllabus 6.2

During electrolysis of aqueous solutions, either the metal cation or hydrogen ions (H^+) from water will be discharged at the cathode.

- (a) Predict whether the metal or hydrogen gas will be discharged at the cathode for each of the following aqueous solutions: [3]

- (i) Aqueous copper(II) nitrate
- (ii) Aqueous magnesium sulfate
- (iii) Aqueous silver nitrate

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- (b) State the general rule relating the reactivity / standard electrode potential of a metal to whether hydrogen or the metal is discharged at the cathode. [2]

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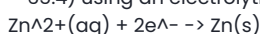
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42 Time Required for Zinc Galvanising — 9701/23/O/N/23/Q5(b)

HARD · Syllabus 6.2

A steel sheet with an area of 500 cm^2 is to be electroplated with a 0.050 mm thick protective layer of zinc (density = 7.14 g cm^{-3} , $A_r = 65.4$) using an electrolytic bath operating at a current of 15.0 A .



- (a) Calculate the mass of zinc required to coat the sheet. [2]

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- (b) Calculate the time, in minutes, required to complete the electroplating process. [3]

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43 Electrolysis of Aqueous Sodium Sulfate with Indicator — 9701/22/F/M/25/Q5(a)-(c)

HARD · Syllabus 6.2

Aqueous sodium sulfate, Na_2SO_4 , containing universal indicator is electrolysed using inert carbon electrodes.

- (a) Write the half-equation for the reaction occurring at the cathode and state the colour observed around the cathode. [2]

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- (b) Write the half-equation for the reaction occurring at the anode and state the colour observed around the anode. [2]

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- (c) Explain the overall chemical change that has occurred to the solution. [1]

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44 Membrane Cell for the Chlor-Alkali Industry — 9701/21/F/M/25/Q5(a)-(b)

HARD · Syllabus 6.2

Modern industrial chlor-alkali plants utilise a polymer ion-exchange membrane cell to separate the anode and cathode compartments.

- (a) State the function of the cation-exchange membrane in this cell. [2]

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- (b) Explain why it is critical to prevent hydroxide ions (OH^-) and chlorine gas (Cl_2) from coming into contact within the electrolytic cell. [2]

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45 Electrolysis of Aqueous Potassium Iodide — 9701/22/M/J/25/Q6(a)-(c)

EASY · Syllabus 6.2

Aqueous potassium iodide, KI , is electrolysed using graphite electrodes in a U-tube.

- (a) State the observations at the cathode and anode. [2]

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- (b) Describe a chemical test on the brown liquid formed around the anode to confirm its identity. [2]

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- (c) Write ionic half-equations for both electrode processes. [2]

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46 Deducing Ionic Charge from Faraday Data — 9701/23/M/J/25/Q5(a)–(b)

HARD · Syllabus 6.2

In an experiment, a current of 1.25 A was passed through an aqueous solution containing an unknown metal ion, M^{z+} , for 38.6 minutes. A mass of 0.805 g of metal M ($A_r = 107.4$) was deposited at the cathode.

- (a) Calculate the amount, in moles, of electrons transferred. [2]

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- (b) Calculate the value of the charge z on the metal ion M^{z+} . [2]

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47 Electrolytic Extraction of Magnesium from Seawater — 9701/21/O/N/25/Q6(a)–(b)

HARD · Syllabus 6.2

Magnesium is produced commercially by the electrolysis of molten anhydrous magnesium chloride, $MgCl_2$, at 750 °C.

- (a) Explain why magnesium cannot be extracted by the electrolysis of aqueous magnesium chloride. [2]

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- (b) Write the half-equation for the cathode reaction in molten $MgCl_2$. [1]

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- (c) Calculate the mass of magnesium ($A_r = 24.3$) produced per hour by an industrial cell operating continuously at 50,000 A. [3]

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48 Current Efficiency in Industrial Electrolysis — 9701/22/O/N/25/Q6(a)–(b)

HARD · Syllabus 6.2

In commercial electroplating, side reactions may reduce the yield of metal deposited:

% Current Efficiency = (Actual mass deposited / Theoretical mass) $\times$ 100%

- (a) A chromium electroplating bath operated at 20.0 A for 2.00 hours, depositing 9.60 g of chromium ($A_r = 52.0$, $Cr^{3+} + 3e^- \rightarrow Cr$). [3]
Calculate the theoretical mass of chromium that should be deposited.

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- (b) Calculate the percentage current efficiency of this electroplating process and suggest one side reaction responsible for inefficiency. [2]

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49 Electrolysis of Aqueous Lead(II) Nitrate — 9701/23/O/N/25/Q5(a)–(b)

HARD · Syllabus 6.2

When aqueous lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2(\text{aq})$, is electrolysed with platinum electrodes, a shiny crystalline metal deposits at the cathode while a dark brown solid deposits at the anode.

- (a) Identify the crystalline metal deposited at the cathode and write its half-equation. [2]

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- (b) The dark brown solid deposited at the anode is lead(IV) oxide, PbO_2 . Write the half-equation for the oxidation of aqueous Pb^{2+} to solid PbO_2 . [2]

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50 Comprehensive Synoptic: Copper Refining, Anode Sludge & Faraday Calculations — 9701/22/M/J/25/Q5(a)–(d)

HARD · Syllabus 6.2

In an industrial copper refinery, an impure blister copper anode containing 98.0% Cu, 0.80% Ag, and 1.20% inert impurities by mass was refined at a constant current of 450 A for 12.0 hours.

(Ar: Cu = 63.5, Ag = 107.9; $F = 96500 \text{ C mol}^{-1}$)

- (a) Calculate the theoretical mass of pure copper deposited on the cathode. [3]

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- (b) Assuming 100% current efficiency for copper deposition, calculate the total mass of blister copper anode dissolved. [2]

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- (c) Calculate the mass of precious silver recovered from the anode slime. [2]

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- (d) Explain why silver does not dissolve and deposit onto the cathode alongside copper. [2]

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Mark Scheme — Topic 6 — Electrochemistry

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	(i) -2 (1); (ii) 0 (1); (iii) +4 (1); (iv) +6 (1)	4
1(b)	$\text{S}_2\text{O}_3^{2-}$: +2 (1); $\text{S}_4\text{O}_6^{2-}$: +2.5 (or +5/2) (1)	2
2(a)	A redox reaction in which atoms of the same element (1); are simultaneously oxidised and reduced (1)	2
2(b)	$\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ (1); Correct balancing and species (1)	2
2(c)	In NaCl: -1 (1); In NaClO: +1 (1)	2
2(d)	$3\text{Cl}_2(\text{g}) + 6\text{NaOH}(\text{aq}) \rightarrow 5\text{NaCl}(\text{aq}) + \text{NaClO}_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$ (1); Oxidation state in NaClO ₃ is +5 (1)	2
3(a)	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ (2)	2
3(b)	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$ (2)	2
3(c)	Orange to green (1)	1
4(a)	From +3 (in $\text{C}_2\text{O}_4^{2-}$) to +4 (in CO ₂) (1)	1
4(b)	$n(\text{H}_2\text{C}_2\text{O}_4) = 0.0500 \times 0.0250 = 1.25 \times 10^{-3} \text{ mol}$ (1); $n(\text{MnO}_4^-) = 1.25 \times 10^{-3} \times (2 / 5) = 5.00 \times 10^{-4} \text{ mol}$ (1); $c(\text{KMnO}_4) = 5.00 \times 10^{-4} / 0.02040 = 0.0245 \text{ mol dm}^{-3}$ (allow 0.0245 – 0.0246 mol dm ⁻³) (1)	3
4(c)	KMnO ₄ acts as a self-indicator; the end-point is marked by the first permanent pale pink colour caused by a tiny excess of unreacted purple MnO ₄ ⁻ ions (1)	1
5(a)	Reducing agent in Stage 1: Iodide ion / I ⁻ (1); Oxidising agent in Stage 2: Iodine / I ₂ (1)	2
5(b)	Starch indicator (added near the end-point when solution is pale straw-yellow) (1); End-point colour change: blue-black to colourless (1)	2
5(c)	$n(\text{S}_2\text{O}_3^{2-}) = 0.100 \times 0.02280 = 2.28 \times 10^{-3} \text{ mol}$; $n(\text{I}_2) = 1.14 \times 10^{-3} \text{ mol} \Rightarrow n(\text{ClO}^- \text{ in } 25 \text{ cm}^3) = 1.14 \times 10^{-3} \text{ mol}$ (1); $n(\text{ClO}^- \text{ in } 250 \text{ cm}^3) = 1.14 \times 10^{-2} \text{ mol}$ (1); Concentration in original 10.0 cm ³ bleach = $1.14 \times 10^{-2} / 0.0100 = 1.14 \text{ mol dm}^{-3}$ (1)	3
6(a)	$2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ (1); Oxygen in H ₂ O ₂ is -1, in H ₂ O is -2 (reduced), in O ₂ is 0 (oxidised) (1)	2
6(b)	$\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (1)	1
6(c)	$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$ (1)	1
7(a)	(i) -3 (1); (ii) -2 (1); (iii) +1 (1); (iv) +4 (1); (v) +5 (1)	5
8(a)	Copper / Cu is oxidised (0 to +2) (1); Nitrogen in HNO ₃ is reduced (+5 to +4 in NO ₂) (1)	2
8(b)	$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$ (2)	2
8(c)	Brown gas / brown fumes evolved (1)	1
9(a)	$2\text{Cu}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq})$ (1)	1
9(b)	Copper(I) (+1) is simultaneously reduced to copper metal (0) (1); and oxidised to copper(II) (+2) (1)	2
10(a)	$\text{SO}_2 + 2\text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$ (2)	2
10(b)	$\text{Cr}_2\text{O}_7^{2-} + 2\text{H}^+ + 3\text{SO}_2 \rightarrow 2\text{Cr}^{3+} + 3\text{SO}_4^{2-} + \text{H}_2\text{O}$ (2)	2
10(c)	Orange to green (1)	1
11(a)	$n(\text{MnO}_4^-) = 0.0100 \times 0.01820 = 1.82 \times 10^{-4} \text{ mol}$ (1); $n(\text{Fe}^{2+}) = 5 \times 1.82 \times 10^{-4} = 9.10 \times 10^{-4} \text{ mol}$ (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
11(b)	Total moles Fe in 250 cm ³ = $9.10 \times 10^{-4} \times 10 = 9.10 \times 10^{-3}$ mol (1); Mass of Fe = $9.10 \times 10^{-3} \times 55.8 = 0.5078$ g (or 508 mg) (1)	2
11(c)	% Fe = $(0.5078 / 1.750) \times 100\% = 29.0\%$ (allow 29.0% - 29.1%) (1)	1
12(a)	$\text{VO}_2^{+} + \text{V} + 2(-2) = +1 \Rightarrow \text{V} = +5$ (1); $\text{VO}_2^{+} + \text{V} + (-2) = +2 \Rightarrow \text{V} = +4$ (1)	2
12(b)	$2\text{VO}_2^{+} + \text{Zn} + 4\text{H}^{+} \rightarrow 2\text{VO}^{2+} + \text{Zn}^{2+} + 2\text{H}_2\text{O}$ (2)	2
13(a)	Oxidising ability decreases down Group 17 ($\text{Cl}_2 > \text{Br}_2 > \text{I}_2$) (1)	1
13(b)	$\text{Cl}_2(\text{aq}) + 2\text{Br}^{-}(\text{aq}) \rightarrow 2\text{Cl}^{-}(\text{aq}) + \text{Br}_2(\text{aq})$ (1); Orange / orange-brown colour formed (1)	2
13(c)	Down the group, atomic radius increases and electron shielding increases (1); The nucleus exerts weaker electrostatic attraction on incoming electrons, making electron capture less favourable (1)	2
14(a)	Sulfur / colloidal sulfur precipitate (1)	1
14(b)	Oxidation state of sulfur in $\text{S}_2\text{O}_3^{2-}$ is +2 (1); In SO_2 , sulfur is +4 (oxidised) (1); In elemental sulfur $\text{S}(\text{s})$, sulfur is 0 (reduced) (1)	3
15(a)	(i) -4 (1); (ii) -2 (1); (iii) 0 (1); (iv) +2 (1); (v) +4 (1)	5
16(a)	$n(\text{MnO}_4^{-}) = 0.0200 \times 0.01850 = 3.70 \times 10^{-4}$ mol (1)	1
16(b)	$n(\text{H}_2\text{O}_2 \text{ in } 25 \text{ cm}^3) = 3.70 \times 10^{-4} \times (5 / 2) = 9.25 \times 10^{-4}$ mol (1); $n(\text{H}_2\text{O}_2 \text{ in } 500 \text{ cm}^3) = 9.25 \times 10^{-4} \times 20 = 1.85 \times 10^{-2}$ mol (1); Concentration in original 20.0 cm ³ = $1.85 \times 10^{-2} / 0.0200 = 0.925$ mol dm ⁻³ (1)	3
17(a)	Mohr's salt has high purity, does not effloresce or lose water of crystallisation easily, and is much less susceptible to aerial oxidation by atmospheric O_2 (1)	1
17(b)	Total moles Mohr's salt = $3.921 / 392.1 = 0.0100$ mol $\Rightarrow n(\text{Fe}^{2+} \text{ in } 25 \text{ cm}^3) = 2.50 \times 10^{-3}$ mol (1); Stoichiometry: $1 \text{ Cr}_2\text{O}_7^{2-} : 6 \text{ Fe}^{2+} \Rightarrow n(\text{Cr}_2\text{O}_7^{2-}) = 2.50 \times 10^{-3} / 6 = 4.167 \times 10^{-4}$ mol (1); Volume = $4.167 \times 10^{-4} / 0.0200 = 0.02083$ dm ³ = 20.8 cm ³ (allow 20.8 - 20.9 cm ³) (1)	3
18(a)	In NaClO: +1 (1); In NaCl: -1 (1); In NaClO ₃ : +5 (1)	3
18(b)	The active oxidising and germicidal agent ClO^{-} is decomposed into chloride and chlorate(V) ions (1)	1
19(a)	Chloride ions, Cl^{-} , are not strong enough reducing agents to reduce concentrated sulfuric acid (1)	1
19(b)	Brown fumes / orange-brown liquid of Br_2 (and choking gas SO_2) (1); $2\text{NaBr} + 3\text{H}_2\text{SO}_4 \rightarrow 2\text{NaHSO}_4 + \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$ (or $2\text{Br}^{-} + \text{H}_2\text{SO}_4 + 2\text{H}^{+} \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$) (1)	2
19(c)	In SO_2 : +4 (1); In S: 0 (1); In H_2S : -2 (1)	3
20(a)	$\text{SO}_3^{2-} + 2\text{OH}^{-} \rightarrow \text{SO}_4^{2-} + \text{H}_2\text{O} + 2\text{e}^{-}$ (2)	2
20(b)	$2\text{MnO}_4^{-} + 3\text{SO}_3^{2-} + \text{H}_2\text{O} \rightarrow 2\text{MnO}_2 + 3\text{SO}_4^{2-} + 2\text{OH}^{-}$ (2)	2
21(a)	HNO_2 : +3 (1); HNO_3 : +5 (1); NO: +2 (1)	3
21(b)	Nitrogen in oxidation state +3 is simultaneously oxidised to +5 and reduced to +2 (1)	1
22(a)	+1 (1)	1
22(b)	$n(\text{S}_2\text{O}_3^{2-}) = 0.100 \times 0.02360 = 2.36 \times 10^{-3}$ mol; $n(\text{I}_2) = 1.18 \times 10^{-3}$ mol; $n(\text{Cu}^{2+} \text{ in } 25 \text{ cm}^3) = 2.36 \times 10^{-3}$ mol (1); Total $n(\text{Cu} \text{ in } 250 \text{ cm}^3) = 2.36 \times 10^{-2}$ mol (1); Mass of Cu = $2.36 \times 10^{-2} \times 63.5 = 1.499$ g (1)	3
22(c)	% Cu = $(1.499 / 2.500) \times 100\% = 60.0\%$ (1)	1
23(a)	(i) -2 (1); (ii) -1 (1); (iii) -1/2 (or -0.5) (1); (iv) +2 (1)	4
23(b)	Fluorine is more electronegative than oxygen (4.0 vs 3.5 on Pauling scale), so fluorine is assigned the negative oxidation state (-1) (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
24(a)	Iodine in two different oxidation states (+5 in IO_3^- and -1 in I^-) (1); react together to form a single product with an intermediate oxidation state (0 in I_2) (1)	2
24(b)	1 IO_3^- (oxidising agent) : 5 I^- (reducing agent) $\Rightarrow$ ratio 1 : 5 (1)	1
25(a)	Blue species: VO^{2+} (vanadyl ion) (1); Violet species: V^{2+} (1)	2
25(b)	$2\text{VO}^{2+} + \text{Zn} + 4\text{H}^+ \rightarrow 2\text{V}^{3+} + \text{Zn}^{2+} + 2\text{H}_2\text{O}$ (2)	2
26(a)	MnO_4^- is a strong enough oxidising agent to oxidise chloride ions, Cl^- , to toxic chlorine gas, Cl_2 (1); $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{Cl}^- \rightarrow 2\text{Mn}^{2+} + 5\text{Cl}_2 + 8\text{H}_2\text{O}$ (1)	2
26(b)	Extra KMnO_4 would be consumed reacting with Cl^- ions in addition to Fe^{2+} (1); This gives an artificially large titre and overestimates the concentration of Fe^{2+} (1)	2
27(a)	Na^+ , Cl^- , H^+ , OH^- (all four required for 2 marks, any two for 1 mark) (2)	2
27(b)	$2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$ (1); In concentrated solution, the high concentration of Cl^- ions favours their discharge over OH^- ions (1)	2
27(c)	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$ (or $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$) (1)	1
27(d)	Sodium hydroxide, $\text{NaOH}(\text{aq})$ (1)	1
28(a)	Anode (+): Impure copper / blister copper (1); Cathode (-): Pure copper sheet (1)	2
28(b)	Anode: $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ (1); Cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ (1)	2
28(c)	Copper dissolves from the anode into the electrolyte at exactly the same rate that copper deposits onto the cathode (1)	1
28(d)	Silver and gold are less reactive / have more positive standard electrode potentials than copper (1); They are not oxidised at the operating voltage and simply drop unreacted to the bottom as the copper anode dissolves (1)	2
29(a)	A molten ionic compound or aqueous solution containing mobile ions (1); that conducts electricity accompanied by chemical decomposition (1)	2
29(b)	In a metallic wire, current is carried by the flow of delocalised mobile electrons without chemical change (1); In an electrolyte, current is carried by the migration of mobile positive and negative ions towards electrodes with chemical reaction (1)	2
30(a)	In solid PbBr_2 , ions are held firmly in fixed lattice positions and cannot move (1); In molten PbBr_2 , the lattice breaks down and Pb^{2+} and Br^- ions are free to move throughout the liquid (1)	2
30(b)	Anode: $2\text{Br}^-(\text{l}) \rightarrow \text{Br}_2(\text{g}) + 2\text{e}^-$ (1); Cathode: $\text{Pb}^{2+}(\text{l}) + 2\text{e}^- \rightarrow \text{Pb}(\text{l})$ (1)	2
30(c)	Anode: Brown / red-brown fumes of bromine gas evolved (1); Cathode: Silvery / grey bead of molten lead metal formed (1)	2
31(a)	Oxygen / O_2 gas (1); $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ (or $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$) (1)	2
31(b)	In dilute solution, OH^- ions are discharged more readily than Cl^- ions (1); In concentrated solution, the concentration of Cl^- ions is vastly higher than OH^- , so Cl^- ions are preferentially discharged to form Cl_2 gas (1)	2
32(a)	$Q = I \times t$ (where I is in Amperes and t is in seconds) (1)	1
32(b)	$t = 45.0 \times 60 = 2700 \text{ s}$; $Q = 2.50 \times 2700 = 6750 \text{ C}$ (2)	2
32(c)	$n(\text{e}^-) = 6750 / 96500 = 0.06995 \text{ mol}$; $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu} \Rightarrow n(\text{Cu}) = 0.06995 / 2 = 0.03497 \text{ mol}$ (1); Mass = $0.03497 \times 63.5 = 2.22 \text{ g}$ (allow 2.21 – 2.23 g) (1)	2
33(a)	$Q = I \times t = 0.400 \times 3600 = 1440 \text{ C}$ (1)	1
33(b)	$n(\text{Ag}) = 1.610 / 107.9 = 0.01492 \text{ mol}$; $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag} \Rightarrow n(\text{e}^-) = 0.01492 \text{ mol}$ (1); $F = Q / n(\text{e}^-) = 1440 / 0.01492 = 96515 \text{ C mol}^{-1}$ (allow 96400 – 96600) (1)	2
33(c)	$L = F / e = 96515 / (1.602 \times 10^{-19})$ (1); $L = 6.02 \times 10^{23} \text{ mol}^{-1}$ (allow 6.01 – $6.03 \times 10^{23} \text{ mol}^{-1}$) (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
34(a)	Oxygen / O ₂ gas (1); 4OH ⁻ (aq) → O ₂ (g) + 2H ₂ O(l) + 4e ⁻ (or 2H ₂ O → O ₂ + 4H ⁺ + 4e ⁻) (1)	2
34(b)	In Cell 1: The blue color fades to colourless because Cu ²⁺ ions are discharged at cathode while OH ⁻ is discharged at anode, forming sulfuric acid (1); In Cell 2: The blue color remains unchanged (1); because Cu ²⁺ ions are removed at the cathode at the exact same rate that Cu dissolves from the copper anode (1)	3
35(a)	Cathode: 2H ⁺ (aq) + 2e ⁻ → H ₂ (g) (1); Anode: 4OH ⁻ (aq) → O ₂ (g) + 2H ₂ O(l) + 4e ⁻ (or 2H ₂ O → O ₂ + 4H ⁺ + 4e ⁻) (1)	2
35(b)	Volume H ₂ (cathode) : Volume O ₂ (anode) = 2 : 1 (1)	1
35(c)	Each mole of O ₂ evolved requires the transfer of 4 moles of electrons (1); The same 4 moles of electrons produces 2 moles of H ₂ gas (since 2e ⁻ per H ₂), giving twice the volume of H ₂ (1)	2
36(a)	SO ₄ ²⁻ < Cl ⁻ < OH ⁻ < I ⁻ (or SO ₄ ²⁻ < OH ⁻ < Cl ⁻ < I ⁻ depending on concentration) (2)	2
36(b)	SO ₄ ²⁻ and NO ₃ ⁻ ions contain central atoms in their maximum oxidation states (+6 and +5) (1); Hydroxide ions (OH ⁻) or water molecules are much more readily oxidised than sulfate or nitrate oxoanions (1)	2
37(a)	Negative terminal (spoon must act as cathode) (1)	1
37(b)	Electrolyte: Aqueous nickel(II) sulfate, NiSO ₄ (aq) (1); Anode: Pure nickel metal bar / sheet (1)	2
37(c)	Ni ²⁺ (aq) + 2e ⁻ → Ni(s) (1)	1
38(a)	Oxidation occurs at the anode (1); Oxygen gas is liberated at the anode, which immediately reacts with the aluminium surface to form an oxide layer (1)	2
38(b)	2Al(s) + 3H ₂ O(l) → Al ₂ O ₃ (s) + 6H ⁺ (aq) + 6e ⁻ (or 4Al + 3O ₂ → 2Al ₂ O ₃) (2)	2
38(c)	Increased corrosion resistance / increased surface hardness / ability to absorb colored dyes (1)	1
39(a)	t = 25.0 x 60 = 1500 s; Q = 3.20 x 1500 = 4800 C (1); n(e ⁻) = 4800 / 96500 = 0.04974 mol (1)	2
39(b)	n(O ₂) = 0.04974 / 4 = 0.01244 mol (1); V = 0.01244 x 24000 cm ³ = 298 cm ³ (allow 297 – 299 cm ³) (1)	2
40(a)	Cryolite lowers the melting point from over 2000 °C to ~950 °C (saving immense energy) (1); Molten cryolite increases the electrical conductivity of the molten electrolyte (1)	2
40(b)	Cathode: Al ³⁺ + 3e ⁻ → Al(l) (1); Anode: 2O ²⁻ → O ₂ (g) + 4e ⁻ (1)	2
40(c)	The oxygen gas produced at high operating temperature (950 °C) reacts with the carbon anodes to form carbon dioxide: C(s) + O ₂ (g) → CO ₂ (g) (1); The carbon anodes gradually burn away and become depleted (1)	2
41(a)	(i) Copper / Cu metal (1); (ii) Hydrogen / H ₂ gas (1); (iii) Silver / Ag metal (1)	3
41(b)	Metals more reactive than hydrogen (more negative standard electrode potential, e.g. Group 1, Group 2, Al) result in H ₂ discharge (1); Metals less reactive than hydrogen (more positive standard electrode potential, e.g. Cu, Ag, Au) are discharged as metal (1)	2
42(a)	Volume = 500 x (0.050 / 10) = 2.50 cm ³ (1); Mass = density x volume = 7.14 x 2.50 = 17.85 g (1)	2
42(b)	n(Zn) = 17.85 / 65.4 = 0.2729 mol; n(e ⁻) = 2 x 0.2729 = 0.5459 mol (1); Q = 0.5459 x 96500 = 52677 C (1); t = Q / I = 52677 / 15.0 = 3512 s = 58.5 minutes (allow 58 – 59 min) (1)	3
43(a)	2H ₂ O(l) + 2e ⁻ → H ₂ (g) + 2OH ⁻ (aq) (1); Purple / blue colour formed due to generation of alkaline OH ⁻ ions (1)	2
43(b)	2H ₂ O(l) → O ₂ (g) + 4H ⁺ (aq) + 4e ⁻ (1); Red / orange colour formed due to generation of acidic H ⁺ ions (1)	2
43(c)	The overall reaction is the electrolysis / decomposition of water (2H ₂ O → 2H ₂ + O ₂); Na ₂ SO ₄ acts merely as an inert electrolyte to conduct current (1)	1
44(a)	The membrane allows Na ⁺ ions to migrate from the anode compartment to the cathode compartment (1); while preventing Cl ⁻ ions and OH ⁻ ions from crossing over (1)	2
44(b)	Cl ₂ reacts rapidly with OH ⁻ to form chlorate(I) ions (ClO ⁻) and chloride ions (1); This contaminates the NaOH product and reduces the yield of chlorine gas (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
45(a)	Cathode: Effervescence / bubbles of colourless gas (H ₂) (1); Anode: Brown liquid / dark precipitate (I ₂) formed (1)	2
45(b)	Add starch solution (1); Turns intense blue-black, confirming iodine (1)	2
45(c)	Cathode: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$ (1); Anode: $2\text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2\text{e}^-$ (1)	2
46(a)	$t = 38.6 \times 60 = 2316 \text{ s}$; $Q = 1.25 \times 2316 = 2895 \text{ C}$ (1); $n(\text{e}^-) = 2895 / 96500 = 0.0300 \text{ mol}$ (1)	2
46(b)	$n(\text{M}) = 0.805 / 107.4 = 7.495 \times 10^{-3} \text{ mol}$ (1); $z = n(\text{e}^-) / n(\text{M}) = 0.0300 / (7.495 \times 10^{-3}) = 4.00 \Rightarrow z = 4$ (ion is M^{4+}) (1)	2
47(a)	In aqueous solution, H^+ ions from water are reduced much more readily than Mg^{2+} ions (1); Hydrogen gas is evolved at the cathode instead of magnesium metal (1)	2
47(b)	$\text{Mg}^{2+}(\text{l}) + 2\text{e}^- \rightarrow \text{Mg}(\text{l})$ (1)	1
47(c)	Q in 1 hour = $50000 \times 3600 = 1.80 \times 10^8 \text{ C}$ (1); $n(\text{e}^-) = (1.80 \times 10^8) / 96500 = 1865.3 \text{ mol}$; $n(\text{Mg}) = 1865.3 / 2 = 932.6 \text{ mol}$ (1); Mass = $932.6 \times 24.3 = 22663 \text{ g} = 22.7 \text{ kg}$ (allow 22.6 – 22.8 kg) (1)	3
48(a)	$t = 2.00 \times 3600 = 7200 \text{ s}$; $Q = 20.0 \times 7200 = 144000 \text{ C}$ (1); $n(\text{e}^-) = 144000 / 96500 = 1.492 \text{ mol}$; $n(\text{Cr}) = 1.492 / 3 = 0.4974 \text{ mol}$ (1); Theoretical mass = $0.4974 \times 52.0 = 25.86 \text{ g}$ (1)	3
48(b)	Efficiency = $(9.60 / 25.86) \times 100\% = 37.1\%$ (allow 37.0% – 37.2%) (1); Side reaction: Simultaneous reduction of H^+ ions forming H_2 gas at the cathode (1)	2
49(a)	Lead / Pb metal (1); $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$ (1)	2
49(b)	$\text{Pb}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{PbO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^-$ (2)	2
50(a)	$t = 12.0 \times 3600 = 43200 \text{ s}$; $Q = 450 \times 43200 = 1.944 \times 10^7 \text{ C}$ (1); $n(\text{e}^-) = (1.944 \times 10^7) / 96500 = 201.45 \text{ mol}$; $n(\text{Cu}) = 201.45 / 2 = 100.73 \text{ mol}$ (1); Mass pure Cu = $100.73 \times 63.5 = 6396 \text{ g} = 6.40 \text{ kg}$ (allow 6.39 – 6.41 kg) (1)	3
50(b)	Mass of Cu dissolved = 6396 g; Blister copper mass = $6396 / 0.980 = 6527 \text{ g} = 6.53 \text{ kg}$ (1); Correct calculation to 3 s.f. (1)	2
50(c)	Mass of silver = $0.0080 \times 6527 \text{ g} = 52.2 \text{ g}$ (allow 52.0 – 52.3 g) (2)	2
50(d)	Silver is less electropositive / has a more positive standard electrode potential than copper (1); At the controlled cell voltage used for copper oxidation, silver cannot be oxidised to Ag^+ and falls unreacted into the anode sludge (1)	2